Q 4.8

1. Proof of mean and covariance:

By the linearity of Expectation:

$$\mathbb{E}[y] = \mathbb{E}[Ax+b] = A\mathbb{E}[x]+b = A\mu_x + b = \mu_y$$

Covariance:

$$\Sigma_y = \mathbb{E}[(y-\mu_y)(y-\mu_y)^T] = \mathbb{E}[(Ax+b-(A\mu_x+b)(Ax+b-(A\mu_x+b))^T] =$$

$$= \mathbb{E}[A(x-\mu_x)(A(x-\mu_x))^T] = A\cdot\mathbb{E}[(x-\mu_x)(x-\mu_x)^T]\cdot A^T =$$

$$= A\cdot\Sigma_x\cdot A^T$$

2. Proof of probability density function:

$y = Ax+b \rightarrow g(x) := Ax+b \rightarrow g^{-1}(y) = x = A^{-1}(y-b)$, $|J_{g^{-1}}| = \frac{1}{|A|}$.

$$f_y(y) = \frac{1}{\sqrt{(2\pi)^n\cdot|\Sigma_x|\cdot|A|^2}}\cdot\exp\left(-\frac{1}{2}\cdot\|A^{-1}(y-b)-\mu_x\|^2_{\Sigma_x}\right)$$

$$\sqrt{|\Sigma_x|\cdot|A|^2} = \sqrt{|A\cdot\Sigma_x\cdot A^T|} = \sqrt{|\Sigma_y|}$$

The inner Mahalahobis term simplifies to: $A^{-1}(y-b)-A^{-1}A\mu_x = A^{-1}(y-\mu_y)$

Applying the Mahalahobis norm property with $M=A^{-1}$, $u=y-\mu_y$:

$$\|A^{-1}(y-\mu_y)\|^2_{\Sigma_x} = \|y-\mu_y\|^2_{(A^{-1})^{-1}\cdot\Sigma_x\cdot(A^{-1})^{-T}} = \|y-\mu_y\|^2_{A\Sigma_xA^T} = \|y-\mu_y\|^2_{\Sigma_y}$$

$$\Rightarrow f_y(y) = \frac{1}{\sqrt{(2\pi)^n\cdot|\Sigma_y|}}\cdot\exp\left(-\frac{1}{2}\cdot\|y-\mu_y\|^2_{\Sigma_y}\right), \text{ Thus } y\sim N(A\mu_x+b, A\Sigma_xA^T)$$